

INFT-1205 – Data Communications & Networking II

Capstone Project

DHCP Documentation

Institution: Durham College

By: Group 1

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Document Overview

This document serves as a brief documentation of the DHCP configurations on the Datacore Solutions network. DHCP pools were configured for the IT VLAN and the HR VLAN.

1. show ip dhcp pool – DatacoreRouter

The DHCP pools for the IT and HR VLANs. Displays the IP address range, the leased addresses, and the excluded addresses.

```
DatacoreRouter#show ip dhcp pool
Pool IT :
  Utilization mark (high/low)      : 100 / 0
  Subnet size (first/next)         : 0 / 0
  Total addresses                   : 126
  Leased addresses                  : 1
  Excluded addresses                : 10
  Pending event                    : none
  1 subnet is currently in the pool :
  Current index      IP address range  Leased/Excluded/Total
  10.160.0.13        10.160.0.1 - 10.160.0.126  1 / 10 / 126

Pool HR :
  Utilization mark (high/low)      : 100 / 0
  Subnet size (first/next)         : 0 / 0
  Total addresses                   : 62
  Leased addresses                  : 1
  Excluded addresses                : 10
  Pending event                    : none
  1 subnet is currently in the pool :
  Current index      IP address range  Leased/Excluded/Total
  10.160.1.12        10.160.1.1 - 10.160.1.62  1 / 10 / 62
DatacoreRouter#
```

2. show ip dhcp binding – DatacoreRouter

Displays the current DHCP leases. FL1 host 1 received an address in the 10.160.0.11 – 10.160.0.126 range, since the first 10 addresses were excluded. FL2 host 1 received an address in the 10.160.1.11 – 10.160.1.62 range, since the first 10 addresses were excluded.

```
DatacoreRouter#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration    Type      State      Interface
ce              Hardware address/
User name
10.160.0.12     0124.fbe3.896c.43  Apr 11 2026 11:28 PM Automatic Active      Gigabit
Ethernet0/0/0.10
10.160.1.12     01c8.4d44.2737.d2  Apr 11 2026 11:44 PM Automatic Active      Gigabit
Ethernet0/0/0.40
DatacoreRouter#
```